PALEOMAGNETIC RESULTS FROM THE ORDOVICIAN MOCCASIN, BAYS, AND CHAPMAN RIDGE FORMATIONS OF THE VALLEY AND RIDGE PROVINCE, EASTERN TENNESSEE

Doyle R. Watts and Rob Van der Voo

Department of Geology and Mineralogy, University of Michigan, Ann Arbor, Michigan 48109

Abstract. New results from the late Early (middle) Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Central Appalachian Mountains indicate that the Early Ordovician was a period of relatively rapid apparent polar wander. The available reliable Late Ordovician poles (Juniata, Beemerville, Trenton) lie in a cluster near 35°N, 120°E separated from the Late Cambrian poles (Wilberns, Lamotte, Wichita Granites) by about 50° of arc. High blocking-temperature components with positive fold tests from the Moccasin and Bays Formations yield a pole position intermediate between the two clusters at 33°N, 147°E (K=135, A_{95}=5.8°, N=6 sites). High blocking-temperature components, also with positive fold tests, from the underlying Chapman Ridge Formation give a pole position that agrees well with the Juniata results, and is about 30° from the Moccasin-Bays pole (27°N, 112°E, K=38.3, A_{95}=15.0°, N=4 sites). At least two interpretations of these data are possible: (1) the Chapman Ridge Formation acquired its characteristic magnetization during the Late Ordovician or Early Silurian sometime after deposition but before its folding and (2) the path of apparent polar wander with respect to North America described a loop during the Ordovician.

Introduction

Van der Voo and French [1977] list three reliable Ordovician paleomagnetic pole positions for stable North America. These include results from the Juniata Formation [Van der Voo and French, 1977], the Beemerville Complex [Proko and Hargraves, 1973] and the Trenton Limestone [McElhinny and Opdyke, 1973]. These poles lie in a cluster near 35°N, 120°E and are separated from the North American Late Cambrian poles by some 50° of arc. In an effort to determine further the nature of the path of apparent polar wander with respect to the North American Craton during the interval spanned by the Late Cambrian through the Ordovician, oriented drill cores were collected from three late Early (middle) Ordovician sedimentary formations cropping out in the Valley and Ridge Province of eastern Tennessee. The units sampled are the Moccasin, Bays, and Chapman Ridge Formations of the Chickamaugua Group.

The Moccasin Formation is described by Rodgers [1953] as a mainly maroon calcareous shale, siltstone, and silty limestone sequence alternating with blue-weathering nonsilty limestone and is exposed in many of the strike belts in eastern Tennessee through Virginia. The Bays Formation is described by Rodgers [1953] as consisting of interfingering calcareous maroon shale, siltstone and silty sandstone, passing into silty and shaley maroon and blue limestone below, and passing into white clear sandstone and fine conglomerate above. The Moccasin is regarded as a more calcareous facies of the Bays Formation by various authors [Rodgers, 1953; Twehhofel et al., 1954]. The outcrops of the Bays Formation are rather restricted and are generally confined to the southeastern portion of Tennessee. The age of the Moccasin and Bays Formations is somewhat uncertain but is near the Blackriverian and Trentonian boundary [Twehhofel et al., 1954], i.e., Llanvirn-Llandeilo.

The Chapman Ridge Formation was defined by Cattermole [1955] to describe medium- to fine-grained calcareous and hematitic gray sandstones in the Knoxville area of Tennessee. The precise status and correlation of the Chapman Ridge Formation is a matter of debate, and the problem has been fully summarized by Milici [1973]. What is important in terms of this study is that the strata sampled, as shown by Cattermole [1958], are clearly middle Ordovician and underlie the Bays Formation with some 1100 feet of Ottosee Shale separating the two units.

The folding and thrusting of the Ordovician strata probably took place during the Late Carboniferous to Permian Alleghenian orogeny. Rodgers [1953, p. 126] states that there is no evidence of earlier, Taconic or Acadian, deformation.

Sampling and Measurement

Figure 1 shows the location of the sampling sites in each formation with respect to the major thrust faults in eastern Tennessee. All samples were collected with a Roc Drill Systems portable coring apparatus and were oriented using a clinometer and Brunton compass. During the sampling of each site, an effort was made to drill the finest-grained, nonfriable beds. Generally, from one to three cores were collected from a given bed, with at least five cores from a given site.

Twenty-eight samples were collected from four sites in the Moccasin Formation. All four sites are on the Copper Creek thrust sheet (Figure 1). Sites 2 through 4 are from upright beds with eastward dips. Site 1 is on the overturned limb of a syncline, just beyond the leading edge of the Saltville Fault (William Brent, personal communication, 1977), providing an excellent fold test for this unit.

A total of 16 samples were collected from two sites in the Bays Formation, just west of the leading edge of the Great Smoky Allochton (Figure 1). The two sites were from outcrops with opposing dips affording the possibility of a fold test in this unit.

The Chapman Ridge Formation was sampled at four sites along the type section defined by

Copyright 1979 by the American Geophysical Union.

Paper number 8B0751.
0148-0227/79/008B-0751$01.00

TABLE 1. Site Mean Directions, Statistics, and Pole Positions

	Site Location, deg	R_s	Before Tilt Correction			After Tilt Correction			α_{95}	Pole Position, deg
			Declination	Inclination	k	Declination	Inclination	k		
			Moccasin, High Blocking-Temperature Directions							
Site 1	36°20'53"N, 83°11'06"W	8/8	181°	55°	84.8	320°	-14°	84.8	6.1	33°N, 146°E
Site 2	36°27'02"N, 83°09'19"W	8/8	318°	-51°	31.0	322°	-10°	31.0	10.1	36°N, 146°E
Site 3	36°25'14"N, 83°13'32"W	2/5	337°	-34°	18.3	331°	-10°	18.3	40.7	40°N, 136°E
Site 4	36°24'23"N, 83°16'40"W	7/7	313°	-44°	39.8	318°	-10°	39.8	9.7	33°N, 150°E
			Moccasin, Intermediate Blocking-Temperature Directions							
Site 1	36°20'53"N, 83°11'06"W	---	---	---	---	---	---	---	---	---
Site 2	36°27'02"N, 83°09'19"W	7/8	168°	48°	24.9	162°	7°	24.9	12.3	47°N, 124°E
Site 3	36°25'14"N, 83°13'32"W	5/5	169°	40°	613	158°	19°	613	3.1°	40°N, 125°E
Site 4	36°24'23"N, 83°16'40"W	7/7	158°	60°	73.8	154°	26°	73.8	7.9°	35°N, 128°E
			Bays, Characteristic Directions							
Site 1	35°33'13"N, 84°05'32"W	11/11	135°	32°	24.6	137°	22°	24.6	9.4°	28°N, 145°E
Site 2	35°34'00"N, 84°06'01"W	5/5	309°	29°	62.2	310°	-10°	62.2	9.8°	28°N, 155°E
			Chapman Ridge, Characteristic Directions							
Site 1	35°55'02"N, 83°57'07"W	2/6	12°	-60°	6.6	0°	-31°	6.6	55.5°	37°N, 97°E
Site 2	35°55'09"N, 83°57'10"W	6/6	19°	-78°	21.5	347°	-48°	21.5	14.8°	24°N, 110°E
Site 3	35°55'35"N, 83°57'22"W	9/10	341°	-11°	3.0	346°	-43°	19.3	12.0	28°N, 112°E
Site 4	35°55'38"N, 83°57'23"W	10/10	73°	-64°	2.2	328°	-50°	30.2	8.9°	17°N, 125°E

R_s is the ratio of the number of samples used in the determination of the mean direction to the number of independently oriented samples collected from a site; k is the estimate of Fisher's precision parameter; and α_{95} is the radius of the circle of 95% confidence around the mean direction [see McElhinny, 1973]. The pole positions are calculated, assuming a coaxial geocentric dipole field, from the directions that have been corrected for tilt.

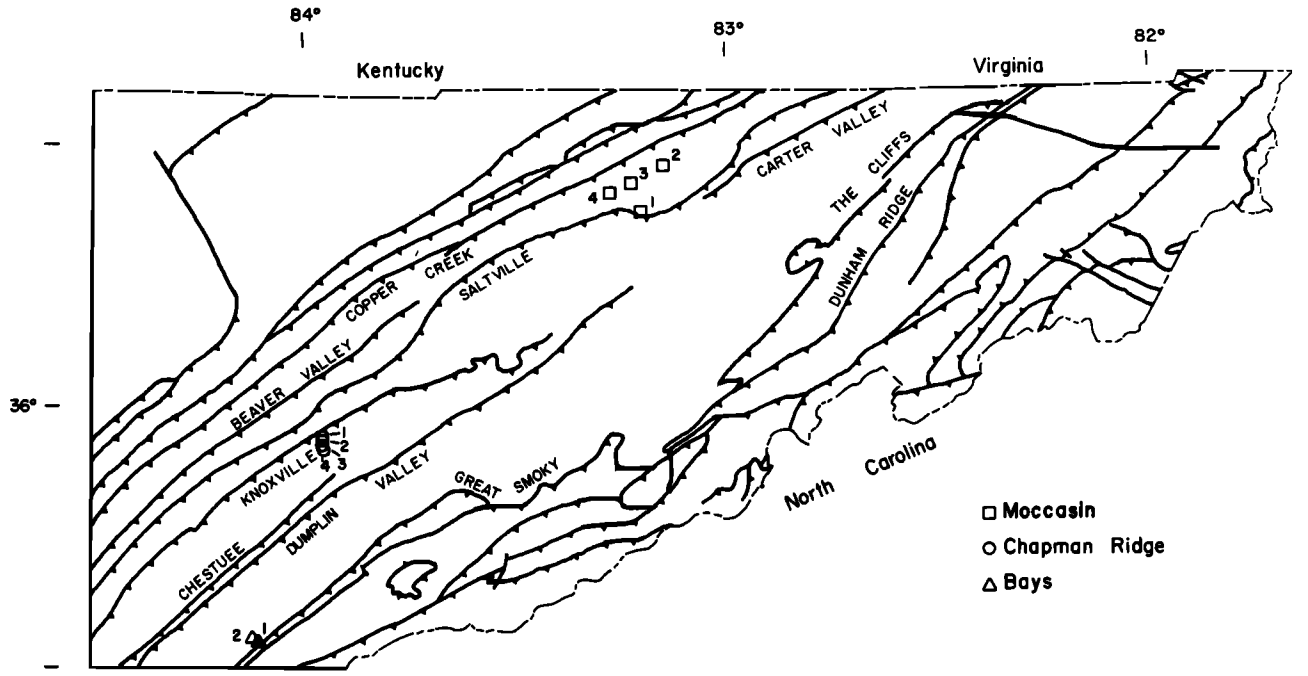


Fig. 1. Map showing site locations for paleomagnetic sampling, in relation to the major thrust sheets of Tennessee's Valley and Ridge Province.

Cattermole [1955]. This section is deformed into numerous small folds with wavelengths ranging from one to tens of meters, providing an excellent opportunity to apply the fold test. One of the folds sampled in this study is pictured in Swingle [1973, p. 71]. Thirty cores were collected from this unit in the medium- to fine-grained beds.

The cores collected from the respective formations were sliced into one or more standard 2.5-cm cylindrical specimens which were measured with a cryogenic magnetometer. At least one specimen from each core was subjected to stepwise thermal demagnetization in a Schonstedt TSD-1 thermal demagnetizing unit. An attempt was made to resolve, as accurately as possible, the components of the natural remanent magnetization of each specimen. After each measurement, orthogonal projections of the magnetization vector [Zijderveld, 1967] were plotted to determine the next demagnetization step that would serve best to isolate the secondary and characteristic directions. The plots were examined for linear trends in the endpoint trajectories defined by three or more endpoints. Characteristic and secondary directions were determined by vector subtraction of two endpoints that best defined the linear trend as determined by visual examination.

Alternating magnetic field demagnetizations were attempted on samples from each unit but resulted in only partial removal of low-coercivity components, the directions of which were ill defined (not having linear trajectories in the orthogonal projections). Chemical demagnetizations were attempted but were not successful because of the high percentage of carbonate cement in the samples investigated.

Table 1 summarizes the site locations, mean directions and associated statistics for the three units.

Magnetization of the Moccasin Formation

The range of behavior exhibited by specimens from the Moccasin Formation during thermal demagnetization is depicted in Figure 2. The projections show the endpoint trajectories corrected for the tilt of the strata. They are interpreted to exhibit two directions of magnetization: (1) a high blocking-temperature component with northwesterly declinations and negative inclinations and (2) an intermediate component with southeasterly declinations and positive inclinations. The four diagrams in Figure 2 illustrate a progressive overprinting of the high blocking-temperature component by the intermediate blocking-temperature component. Figure 2a shows a very stable, nearly univectorial magnetization that persists up to the highest (665°C) temperatures of demagnetization. Figures 2b-2d depict intermediate blocking-temperature components that are fully eliminated at temperatures between 630°C and 650°C, leaving a high blocking-temperature component that persists even in specimens that are nearly completely overprinted.

The high and intermediate blocking-temperature components were isolated using the vector subtraction method described above. The components so isolated are plotted in the equal-area projections in Figure 3 showing the directions before and after structural correction. The high blocking-temperature directions beautifully pass the unfolding test. The in situ directions which plot in the southern quadrants of Figure 3a are all from site 1 which was located in the overturned limb of a syncline. No intermediate blocking-temperature magnetizations were clearly evident in the samples from this critical site. All of the specimens from site 1 exhibited the behavior represented by Figure 2a, i.e., univector directions with high stability upon thermal demagnetization.

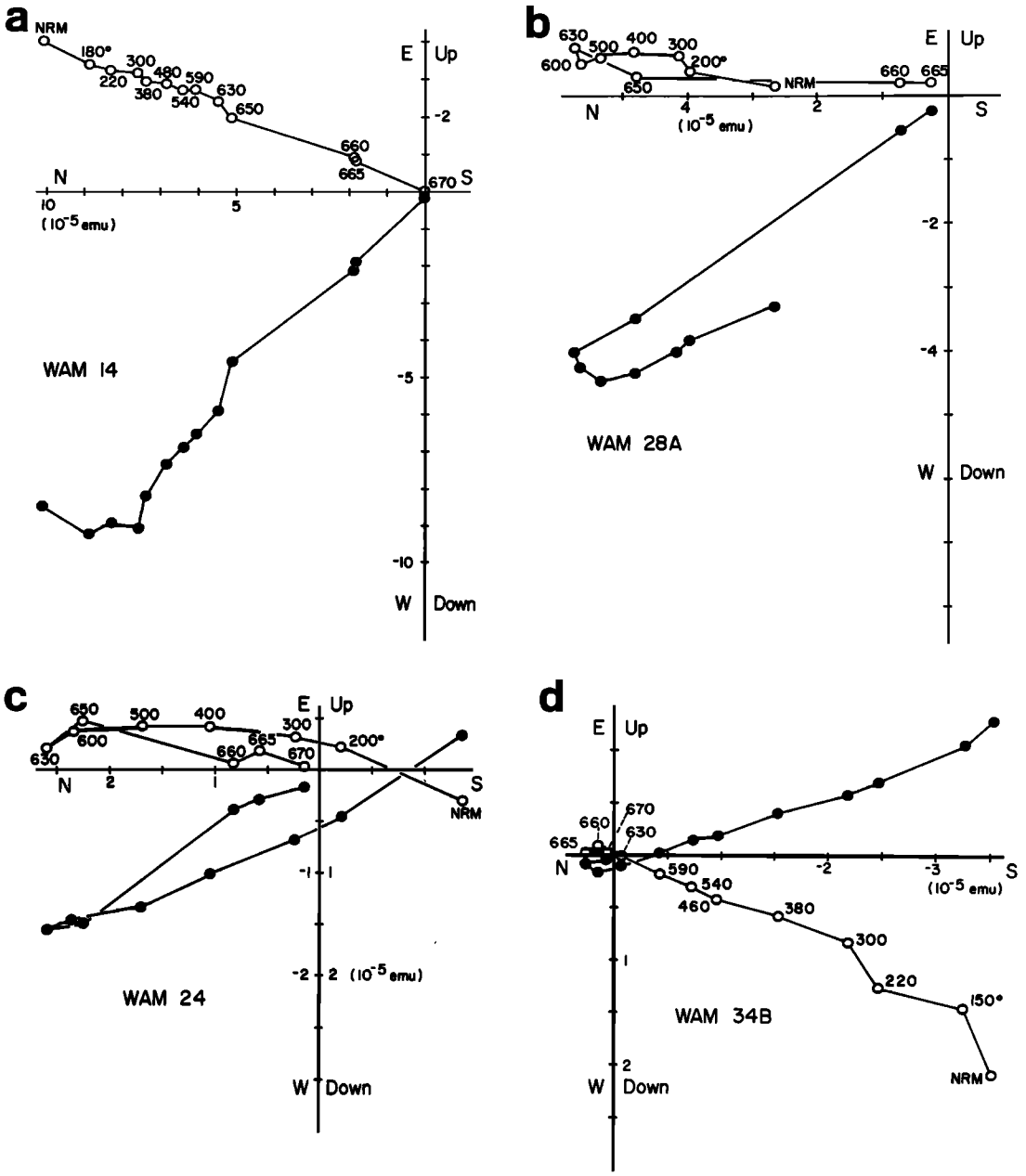


Fig. 2. Orthogonal demagnetization projections [Zijderveld, 1967] of the successive endpoints of the magnetization vector during progressive thermal demagnetization, after correction for the tilt of the strata. Demagnetization temperatures are indicated in degrees Celsius. Solid circles represent projections on the horizontal plane, and open symbols those on the north-south vertical plane. Units indicated for the axes are in 10^{-3} A m^{-1} (10^{-5} emu per sample of 10 cm^3). The samples are from the Moccasin Formation and show from (a) to (d) an increasing amount of overprinting by an intermediate blocking-temperature magnetization.

An ambiguity arises from the geometry of the structural relations with respect to the magnetic directions. As seen by comparing Figure 3a with Figure 3c, the in situ directions of high blocking temperature of site 1 agree rather well with the in situ intermediate blocking-temperature directions of the other sites. If the intermediate directions were acquired after the folding of the Moccasin Formation, they would be nearly parallel to the in situ high blocking-temperature directions in this site and would not be revealed by vector subtraction methods. We note therefore that we distinguished the high-blocking temperature directions from site 1 from the intermediate blocking temperature directions of the other sites by their persistence at high temperatures and not necessarily by their directions.

The significance of the fold test for each unit was determined using the statistical test described by McElhinny [1964]. The estimate k of the precision parameter of the site mean directions was computed before and after structural correction defining a k_b and a k_a respec-

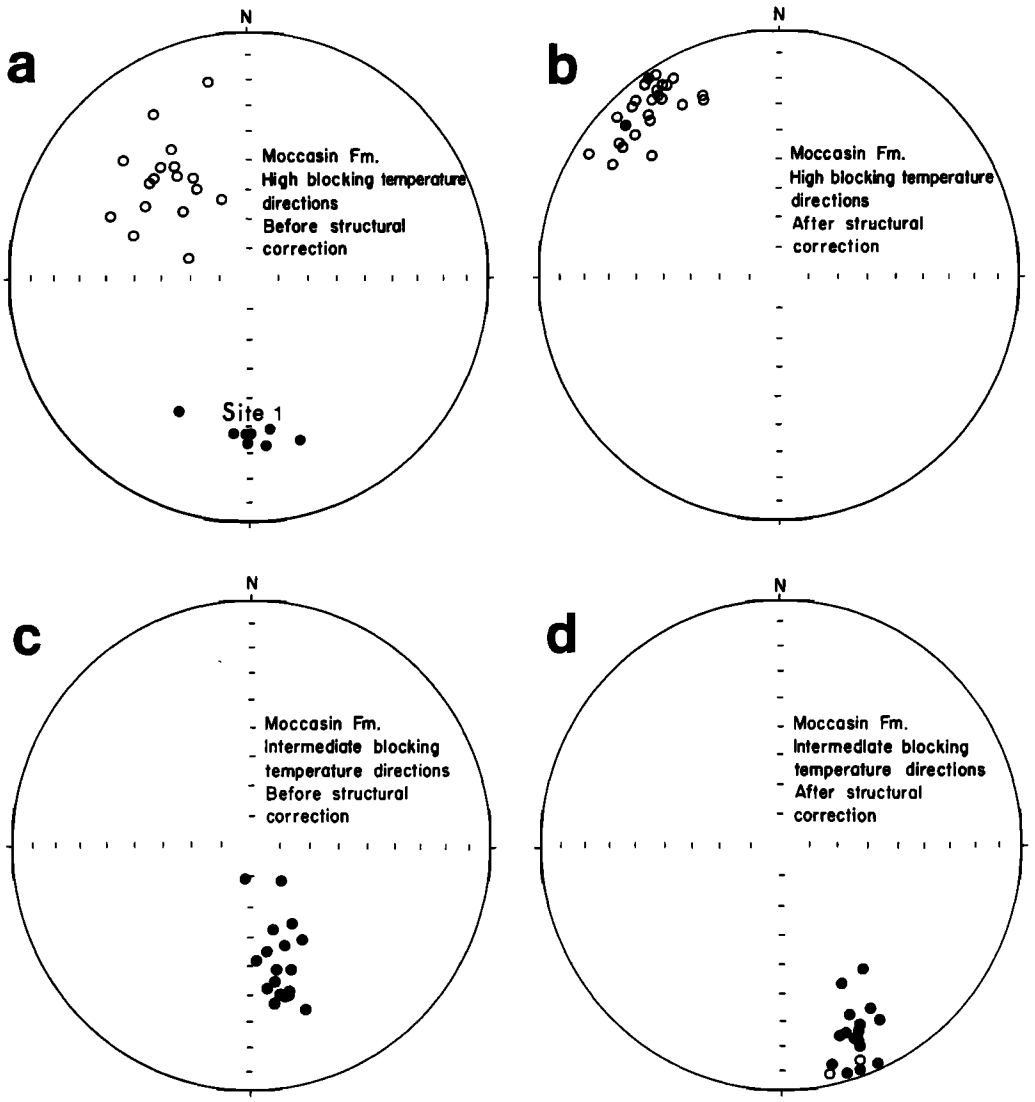


Fig. 3. Equal-area projections of the directions obtained from the samples of the Moccasin Formation. Open (solid) symbols represent projections on the upper (lower) hemisphere.

tively. The ratio, k_a/k_b, was compared to Table 1 in McElhinny [1964] to determine the significance of the fold test. These results are given in Table 2 showing that the fold test for the Moccasin high blocking-temperature directions is positive and significant to the 99% level of confidence.

Contrary to the case of the high blocking-temperature directions, there is no conclusive fold test for the intermediate components (see Table 2), so the timing of the acquisition of these directions is uncertain. The pole obtained from the intermediate in situ directions does not correspond to any postfolding (post-Mississippian) pole positions for stable North America. Therefore a prefolding acquisition of the intermediate directions is favored.

Even though the timing of the acquisition of the high blocking-temperature magnetic components relative to the intermediate blocking-temperature components cannot be conclusively determined, it is assumed that the high blocking-temperature components correspond to the initial, or primary, magnetization and that the intermediate blocking-temperature directions correspond to a secondary magnetization.

Magnetization of the Bays Formation

Only two sites were sampled in the Bays Formation, but the sites were from outcrops with opposite dips so an unfolding test may be applied.

The representative behavior of the Bays Formation samples is depicted in Figure 4a. Stepwise thermal demagnetization revealed high blocking-temperature components characteristic of the unit. Secondary magnetizations are present but are not well resolved. The characteristic directions persist up to the highest demagnetization temperatures, and no evidence for further high-temperature components was detected.

The characteristic directions isolated by vector subtraction are plotted on the equal-area projections in Figure 5, showing the directions before and after structural correction. The directions from site 1 have southeasterly declinations, and the directions from site 2 all have northwesterly declinations. It is apparent that

TABLE 2. Statistical Significance of the Fold Tests (Mean Site Directions)

	N	K	$\bar{D}$	$\bar{I}$	Comments
Moccasin, high blocking-temperature directions					
Before correction	4	1.6	312°	-35°	k_a/k_b^* = 114.8; fold test is significant
After correction	4	183.7	323°	-11°	to the 99% confidence level.
Moccasin, intermediate blocking-temperature directions					
Before correction	3	57.3	166°	49°	k_a/k_b = 1.1; fold test is not signifi-
After correction	3	61.6	158°	17°	cant to the 95% confidence level.
Bays, characteristic directions					
Before correction	2	3.8	312°+	-2°	k_a/k_b = 18.0; fold test is not signifi-
After correction	2	69.5	313°+	-16°	cant to the 95% confidence level.
Moccasin, high blocking temperature-Bays, characteristic directions					
Before correction	6	2.1	312°+	-20°	k_a/k_b = 44.6; fold test is significant
After correction	6	69.5	313°+	-16°	to the 99% confidence level.
Chapman Ridge, characteristic directions					
Before correction	4	5.2	9°	-59°	k_a/k_b = 7.7; fold test is significant to
After correction	4	40.2	346°	-44°	the 95% confidence level but not the 99% confidence level.

*k_b = k before structural correction, k_a = k after structural correction.

+All directions taken as normal polarity in scatter calculation.

structural correction brings the directions from the two sites into nearly antipodal positions. Regarding these as a set of normal and reversed directions, the k before and after correction was calculated, taking all of the directions as normal polarity. Although the k increased from 3.8 to 69.5, this increase is not significant to the 95% confidence level because it represents results

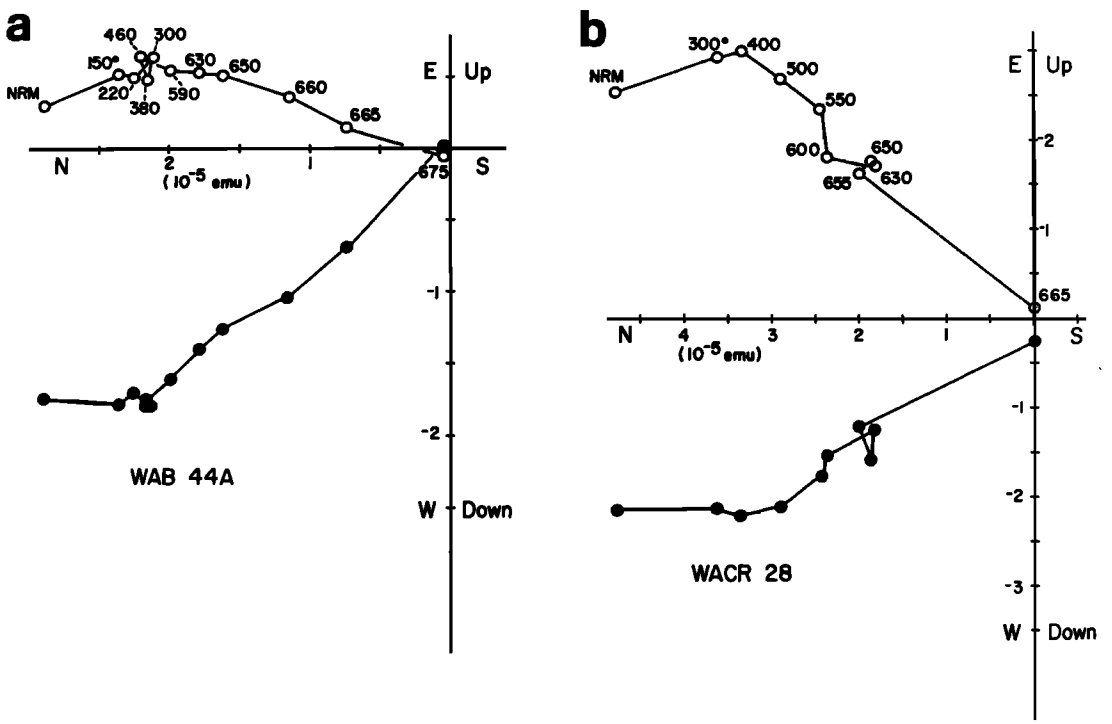


Fig. 4. Orthogonal demagnetization projections of samples from the (a) Bays Formation and (b) Chapman Ridge Formation, after correction for the tilt of the strata. Symbols and units for the axes are the same as those used in Figure 2.

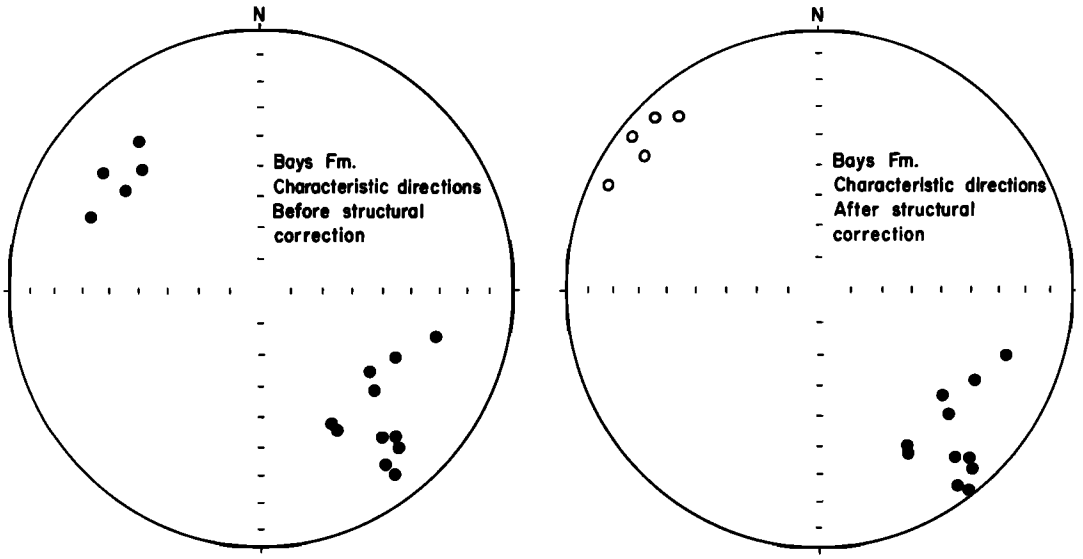


Fig. 5. Equal-area projection of the directions obtained from the samples of the Bays Formation. Symbols are the same as those used in Figure 3.

from only two sites (see Table 2). The results from the Bays Formation alone do not constitute a significant fold test. However, when averaged with the high blocking-temperature directions from the contemporaneous Moccasin Formation, the combined Moccasin-Bays site mean directions yield a positive fold test, significant to the 99% level of confidence (see Table 2).

Magnetization of the Chapman Ridge Formation

The behavior of specimens from the Chapman Ridge Formation during thermal demagnetization was less ideal, perhaps due to the medium-grained texture of many of the specimens analyzed. It was not always possible to isolate a characteristic magnetic direction defined by three or more co-linear points from the orthogonal projections. However, characteristic magnetic directions were isolated from the majority of specimens analyzed, especially from sites 2, 3, and 4. Figure 4b shows a representative behavior of acceptable specimens upon demagnetization. The characteristic direction is revealed at temperatures above 500°C. In many specimens, this characteristic direction does not decay uniformly to the origin, perhaps suggesting the presence of a high blocking-temperature component that is not fully resolved.

The problem in the demagnetization analysis of the Chapman Ridge Formation, alluded to above, was that the specimens acquired large viscous magnetic moments during the time between removal from the demagnetization device to the insertion into the measuring instrument. This problem was especially pronounced after heating above 600°C. The magnetic moment acquired was very unstable, decaying rapidly once the sample was in a field-free space. Therefore a measurement of the magnetic moment for a given specimen was recorded only after there was no significant change with time. The success rate of this procedure could be judged from the linearity of the endpoint trajectories above demagnetization temperatures of 500°C.

Despite these problems in the measurement of the samples, a characteristic component of magnetization could be calculated for most specimens. Secondary magnetizations were sometimes evident but were not well resolved.

Figure 6 is an equal-area projection of the characteristic magnetic directions from the Chapman Ridge Formation. Two within-site fold tests were possible within this unit (site 3 and site 4, see Table 1) and were positive to the 99% level of confidence. Table 2 shows that the mean site directions also give a positive fold test to the 95% level of confidence.

Pole Positions

Figure 7 is an equal-area projection of the site mean directions for the magnetic directions isolated from the Moccasin, Bays, and Chapman Ridge Formations after structural correction has been applied. This diagram shows the agreement between the selected characteristic directions of the Moccasin and the Bays Formations, as well as the antipodal nature of the site 1 directions of the Bays Formation (see also Table 1). The stability of these directions up to the highest demagnetization temperatures and the presence of normal and reversed polarities passing fold tests suggest that the high blocking-temperature directions of the Moccasin Formation and the characteristic directions of the Bays Formation are representative of the middle Ordovician paleomagnetic field with respect to North America. The formation mean directions, their associated statistical parameters, and their pole positions are given in Tables 3a and 3b. The pole positions, furthermore, are given in Table 4 and in Figure 8 along with reliable Silurian, Ordovician, and Late Cambrian pole positions [Van der Voo and French, 1977]. The combined Moccasin-Bays pole occupies a place intermediate between the cluster of poles near 35°N, 120°E and the Late Cambrian poles.

The origin and time of acquisition of the Moccasin intermediate blocking-temperature directions is unresolved. Because of the lower demagnetization temperatures required for their remov-

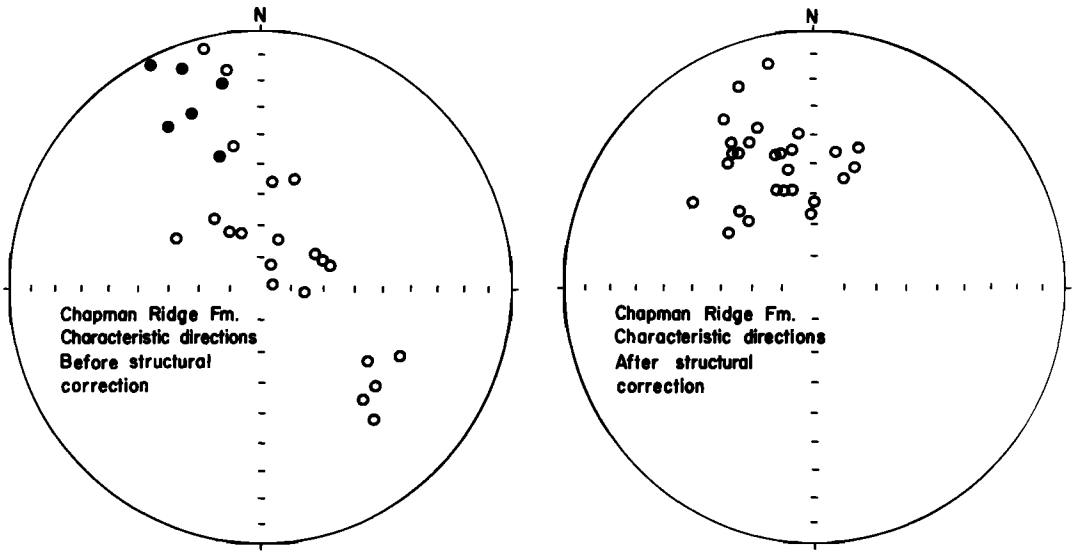


Fig. 6. Equal-area projection of the directions obtained from the samples of the Chapman Ridge Formation. Symbols are the same as those used in Figure 3.

al, and because they are apparently not present at every site (if they are present in site 1, then they are definitely secondary), it is assumed that these directions are secondary. However, as can be deduced from Table 1, the in situ mean directions for the secondary magnetizations do not correspond to any post-Mississippian pole positions for the North American Craton. After structural correction, the secondary pole corresponds to the Late Carboniferous pole positions [Van der Voo and French, 1974] or, possibly, to the Trenton Limestone pole. Thus the geometry of the polar wander path for North America as presently known does not allow the determination of the time of acquisition of the Moccasin secondary magnetization. These directions may indeed be as old as Ordovician and related to the diagenesis of the Moccasin Formation. It is doubtful that the secondary magnetization represents a thermal overprinting because the directions are not ubiquitously present in the area.

As seen in Figure 7 and Table 1, the Chapman Ridge site mean directions are clearly distinguishable from any of the Moccasin or Bays directions, having steeper (more negative) inclinations and more northerly declinations. Figure 8 shows that the Chapman Ridge pole falls in the group of poles near the Late Ordovician Juniata pole. Considering the relative ages of the Moccasin-Bays and Chapman Ridge Formations, there are at least three possible explanations: (1) the Chapman Ridge Formation acquired its characteristic magnetic direction after the overlying Moccasin and Bays Formations during the Upper Ordovician or Silurian through a chemical remagnetization, (2) the Chapman Ridge Formation acquired its characteristic magnetization before the Moccasin and Bays Formations, requiring a loop in the North American polar wander path going from the Chapman Ridge pole, through the poles of the Moccasin and Bays, to the Trenton and Juniata poles, and (3) there have been displacements of the Chapman Ridge Formation and the Moccasin and Bays Formations relative to each other, or relative to the North American craton. However, a relative rotation between the thrust sheets is not favored because of the excellent agreement of the Moccasin and Bays high blocking-temperature directions even though they were sampled from different thrust sheets. Also, relative rotations would affect only the declinations and not the inclinations (which differ by 30°). Of the first two explanations, we tentatively favor the first.

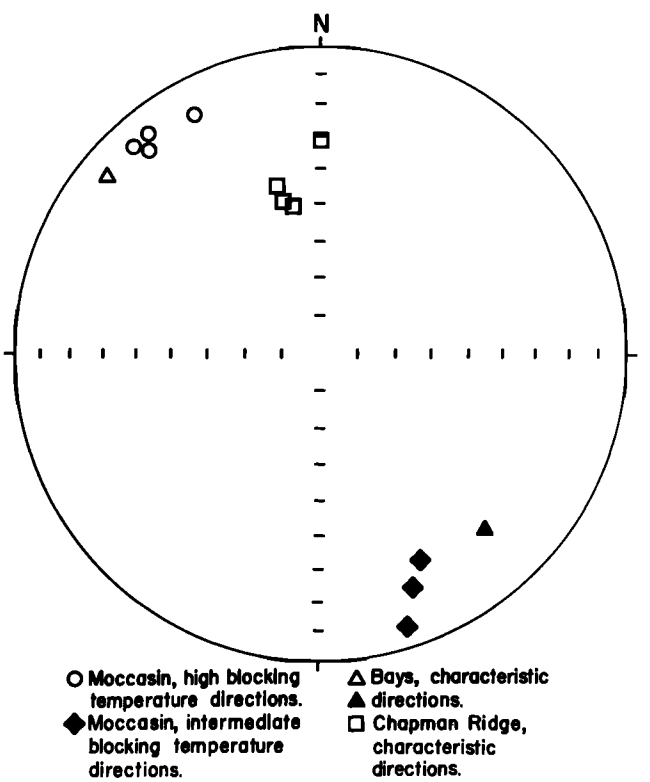


Fig. 7. Equal-area projection of the site mean directions of this investigation. Open (solid) symbols represent projections on the upper (lower) hemisphere, after correction for the tilt of the strata.

TABLE 3a. Mean Formation Pole Positions

	n	Declination	Inclination	k	α_{95}	Position, deg	dp	dm
Moccasin, high-blocking-temperature directions	25	321°	-11°	42.0	4.5°	34°N, 146°E	2.3°	4.6°
Moccasin, intermediate blocking-temperature directions	18	158°	16°	32.7	6.1°	41°N, 125°E	3.2°	6.3°
Bays, characteristic directions	16	314°	-18°	26.7	7.3°	28°N, 148°E	3.9°	7.6°
Chapman Ridge, characteristic directions	27	341°	-46°	18.7	6.6°	24°N, 115°E	5.4°	8.4°

n is the number of samples. k is the estimate of Fisher's precision parameter, and α_{95} is the radius of the circle of 95% confidence around the mean direction [McElhinny, 1973]. dp and dm are the semiaxes of the oval of 95% confidence around the pole position.

TABLE 3b. Mean Pole Positions of the Site Mean Poles

Unit	N	K	α_{95}	Position, deg
Moccasin, high blocking-temperature directions	4	194	6.6°	36°N, 145°E
Moccasin, intermediate blocking-temperature directions	3	169	9.5°	41°N, 126°E
Bays, characteristic directions	2	169	19.4°	28°N, 150°E
Moccasin, high blocking-temperature directions plus Bays characteristic directions	6	135	5.8°	33°N, 147°E
Chapman Ridge, characteristic directions	4	38.3	15.0°	27°N, 112°E

N is the number of site mean poles. K and α_{95} are the estimate of Fisher's precision parameter for the distribution of the site mean poles and the radius of the circle of 95% confidence, respectively [McElhinny, 1973].

TABLE 4. Paleomagnetic Pole Positions (Plotted in Figure 8)

Number	Unit	Age	Pole Position	Reference
9	Bloomsburg Formation	Late Silurian	32°N, 102°E	Roy et al. [1967]
8	Castanea Formation	Early Silurian	21°N, 105°E	quoted in McElhinny and Opdyke [1973]
7	Juniata Formation	Late Ordovician	32°N, 114°E	Van der Voo and French [1977]
6	Beemerville Complex	Late Ordovician	35°N, 126°E	Proko and Hargraves [1973]
5	Trenton Limestone	middle to Late Ordovician	36°N, 116°E	McElhinny and Opdyke [1973]
-	Combined Moccasin and Bays Formations	middle Ordovician	33°N, 147°E	This paper
-	Chapman Ridge Formation	middle Ordovician	27°N, 112°E	This paper
4	Colorado Intrusives	Cambro-Ordovician	37°N, 122°E	Larson and Mutschler [1971]
3	Wilberns Formation	Late Cambrian	6°N, 159°E	Van der Voo et al. [1976]
2	Lamotte Formation	Late Cambrian	1°N, 168°E	Al-Khafaji and Vincenz [1971]; thermal demagnetization
1	Wichita Granites	Late Cambrian	4°N, 164°E	Spall [1970]; thermal demagnetization

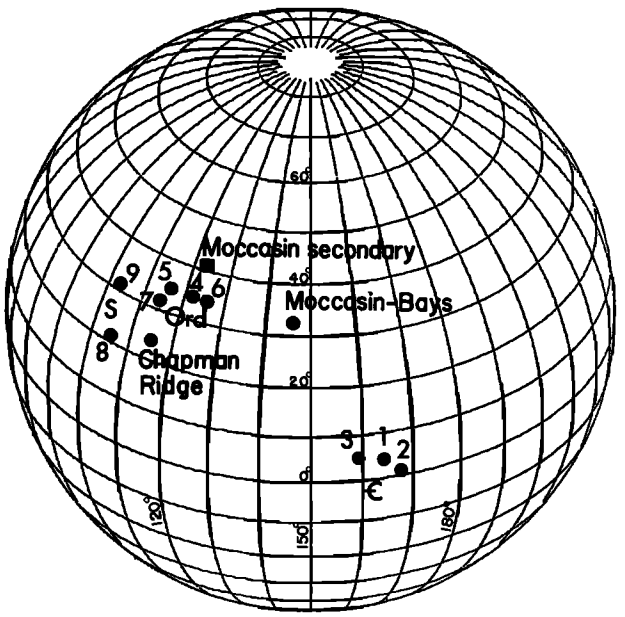


Fig. 8. Plot of pole positions relevant to this study. The numbers of previously published Early Paleozoic poles correspond to those in Table 3. ϵ = Cambrian, Ord = Ordovician, S = Silurian pole positions for the North American craton.

Conclusions

Stepwise thermal demagnetization of oriented samples from the middle Ordovician Moccasin, Bays, and Chapman Ridge Formations reveal high blocking-temperature directions characteristic of the respective units. The characteristic directions all cluster significantly after correction for structural tilting. The characteristic directions of the contemporaneous Moccasin and Bays Formations agree very well and indicate a pole position intermediate between the Late Cambrian poles and the cluster of previously published Ordovician poles. The characteristic directions from the Chapman Ridge Formation give a pole located in the cluster of poles at 35°N, 120°E, agreeing very well with the results from the Upper Ordovician Juniata Formation. This may be the result of a remagnetization of the Chapman Ridge Formation or perhaps a hitherto undefined loop occurs in the Ordovician North American polar wander path. Relative rotations between the various sampling sites are a rather remote possibility.

A secondary magnetization of the Moccasin Formation is revealed by intermediate blocking-temperature directions in many of the specimens. Unfortunately, the fold test is not conclusive for these components because of their absence from a critical outcrop. The pole position obtained from the in situ directions does not correspond to any known North American post-Mississippian pole position, whereas the pole corresponding to the tilt-corrected directions lies near the Carboniferous poles and the Ordovician Trenton pole for North America. Because of the indeterminancy of the age of this magnetization, any discussion on its origin must be speculative. The favored interpretation is that it is a chemical remagnetization, possibly associated with diagenesis, acquired before the folding of the Moccasin Formation.

Acknowledgments. The authors wish to thank William Brent of the Tennessee Division of Geology, Knoxville, Tennessee, for accompanying one of us (D.R.W.) in the field and providing outcrop locations and structural and stratigraphic interpretations. Stuart Maher of the Tennessee Division of Geology and Robert Laurence of the United States Geological Survey suggested promising units for study and provided outcrop location. Eric Inglis assisted in the field. This research was supported by the Division of Earth Sciences, National Science Foundation, NSF grant EAR 76-14996.

References

Al-Khafaji, S. A., and S. A. Vincenz, Magnetization of the Cambrian Lamotte Formation in Missouri, Geophys. J. Roy. Astron. Soc., 24, 175-202, 1971.

Cattermole, J. M., Geology of the Shooks Gap Quadrangle, Tennessee, Map GQ 76, U. S. Geol. Surv., Washington, D. C., 1955.

Larson, E. E., and F. E. Mutschler, Anamalous paleomagnetic pole from isotopically dated Cambro-Ordovician intrusives in Colorado, Geol. Soc. Amer. Bull., 82, 1657-1966, 1971.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, Geophys. J. Roy. Astron. Soc., 8, 338-340, 1964.

McElhinny, M. W., Paleomagnetism and Plate Tectonics, 358 pp., Cambridge University Press, New York, 1973.

McElhinny, M. W., and N. D. Opdyke, Remagnetization hypothesis discounted: A paleomagnetic study of the Trenton limestone, New York State, Geol. Soc. Amer. Bull., 84, 3679-3708, 1973.

Milici, R. C., The stratigraphy of Knox County, Tennessee, Geology of Knox County, Bull. 70, pp, 9-24, Tenn. Div. of Geol., Nashville, 1973.

Proko, M. S., and R. B. Hargraves, Paleomagnetism of the Beemerville (New Jersey) alkaline complex, Geology, 1, 185-186, 1973.

Rodgers, J., Geologic map of East Tennessee with explanatory text, Bull. 58, 168 pp., Tenn. Div. of Geol., Nashville, 1953.

Roy, J. L., N. D. Opdyke, and E. Irving, Further paleomagnetic results from the Bloomsburg Formation, J. Geophys. Res., 72, 5075-5086, 1967.

Spall, H., Paleomagnetism of basement granites in southern Oklahoma, Final report, Okla. Geol. Notes, 30, 136-150, 1970.

Swingle, G. D., Structural geology of Knox County, Tennessee, Geology of Knox County, Bull. 70, pp. 63-73, Tenn. Div. of Geol., Nashville, 1973.

Twehhofel, W. H., et al., Correlation of the Ordovician formations of North America, Geol. Soc. Amer. Bull., 65, 247-298, 1954.

Van der Voo, R., and R. B. French, Apparent polar wander for the Atlantic-bordering continents: Late Carboniferous to Eocene, Earth Sci. Rev., 10, 99-119, 1974.

Van der Voo, R., and R. B. French, Paleomagnetism

of the Late Ordovician Juniata Formation and the remagnetization hypotheses, J. Geophys. Res., 82, 5796-5802, 1977.

Van der Voo, R., R. B. French and D. W. Williams, Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America, J. Geophys. Res., 81, 5633-5638, 1976.

Zijderveld, J. D. A., A.C. demagnetization of rocks: Analysis of results, in Methods in Paleomagnetism, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

(Received February 7, 1978;
revised May 9, 1978;
accepted July 19, 1978.)